Bistatic SAR Radar for Long-Term Snow Pack Monitoring

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***Abstract*— This article presents a short-range, bistatic radar system for *in situ*, long-term snow monitoring. The proposed system is capable of autonomously capturing time-lapsed, 2-D images of the snow cover. In this article, we report on system concept and initial results for average snow-water-equivalent (SWE) assessment. Average SWE of dry snow is estimated based on two-way-travel time, bistatic radar measurements and compared to observed snow pit data as well as monostatic radar estimates. Although current empirical results are limited, measurements are indicating good performance. The bistatic radar instrument is capable of 2-D imaging and wet snow assessment, but empirical validation is required.**

***Index Terms*— Ground penetrating radar, synthetic aperture radar, ultra wideband radar.**

I. Introduction

SNOW pack monitoring instrumentation is currently quite limited and for most variables requires manual expert inspection. This procedure is tedious and costly requiring manual excavation of snow pits from the snow surface down to ground. Each snow pit must accommodate a human observer to make an inspection of the snow pack structure as a function of snow depth. As a minimum, the observer should measure density profiles in order to measure snow-water-equivalence (SWE). Manual observations are limited both in observation frequency as well as in total number of observations. While numerous robust manual observation programs do exist (e.g., regobs [1] in Norway, SLF [2] in the Alps and Natural Resources Conservation Service (NRCS) Snow Course [3] in the U.S.), demanding observation efforts are limiting assessment of snow pack variability on a time scale concurrent with weather forcing variables. In order to improve snow pack modeling, improved instrumentation allowing for frequent observation of snow pack processes is required.

Sustained snow pack monitoring requires self-contained instrumentation for remote, unattended operation with minimal interference of the natural environment. SWE measurements are available with several sensor systems such as the "SnowFox" [4] and Cambpell's Gamma sensor [5]. However, both of these systems are expensive. Other technological solutions for minimal invasive snow measurement like the Swiss Snow Micro Pen (SMP) exists as well [6], still requiring manual operation for repeated measurements.

Mechanical technologies like the snow pillow have been used for point-based SWE measurements. The hydrostatic pressure is monitored over time with a pressure-sensitive pillow on the ground. However, concern about leakage of antifreeze with environmental impact has been raised [7] and no structural information of the snowpack is provided. For additional sensing functionality like imaging, snowpack penetration technologies like radio waves are required.

Radar-based snow assessments have been demonstrated to provide SWE, depth, and stratigraphy estimates continuously. Downward looking frequency-modulated continuous wave (FMCW) radar was pioneered in the seventies [8], [9]. A review of FMCW radar studies in snow was published in 2008 [10] highlighting significant contributions. In 2003 [11], a downward looking multi-static radar systems enabling velocity picking was proposed.

The potential for radar-based snow monitoring is significant and several upward looking radar systems (or UpGPR) for snow assessment are reported [12]–[14] and [15]. However, these systems are limited to just one dimension (1-D) in space and are fairly bulky, costly, and power intensive equipment.

Furthermore, single radar snow observations require additional snow depth information for determining properties like SWE. Depth information can easily be retrieved with a tower-based downward looking radar, as the mounting height is known. However, for an upward looking radar, no absolute references are kept and the geometry is changing with the snow cover.

SAR tomography can be used to reconstruct the permittivity and identify multiple internal layers within the snowpack [16], as well as [17], [18]. However, airborne SAR campaigns are costly. Compact snow radar sensors for airborne remote snowpack sensing is reported in [19], the system is downward looking with good snow surface resolution. Penetration into the snowpack is not reported and is assumed to be minimal in wet snow unless significant power is transmitted. In addition, repeated observations are expensive and infrequent requiring airborne carrier.

Manuscript received April 6, 2018; revised October 12, 2018 and April 26, 2019; accepted August 10, 2019. Date of publication September 16, 2019; date of current version December 27, 2019. This work was supported by the Land-ATmosphere Interactions in Cold Environments (LATICE) Project. LATICE is a strategic research initiative funded by the Faculty of Mathematics and Natural Sciences. *(Corresponding author: J. Håvard H. Eriksrød.)*

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Digital Object Identifier 10.1109/TGRS.2019.2935781

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An additional issue is instrumentation interference of the snow pack under observation. Upward looking radar systems may be buried in ground during the summer season with minimal interference above ground [14]. However, electrically powered instruments will unavoidably dissipate heat and may melt the covering snow. Alternatively, a downward looking radar requires some sort of tower above ground affecting snow coverage depending on size and wind interference. These tradeoffs are still unsettled and must be further explored. However, modern technology trends are toward smaller and power-lean systems with significant potential for minimal measurement interference.

This article proposes a bistatic radar system using power efficient technology for long-term snow pack monitoring. The system is power efficient, fully autonomous, and enables snow assessment without any additional sensors. The proposed system explores a novel, ultrawideband (UWB) radar module with a frequency range spanning from 2.75 to 5.25 GHz for continuous measurement of snow properties carefully designed for unattended, remote operation.

In this article, we explore a coherent, pulsed radar system for bulk SWE monitoring. The bistatic SAR radar is a versatile instrument with additional possibilities. As indicated in [20], dry snow packs are primary sensed by signal propagation delay (permittivity), while wet snow is affecting the signal attenuation. Coherent radar systems enable measurement of both permittivity and signal loss (loss tangent) enabling assessment of both wet and dry snowpacks provided sufficient radar sensitivity. Measurements like snow pack stratigraphy are feasible with adequate signal processing and signal loss analysis may be used for assessment of wet snow as well as precipitation during the summer season.

This article is organized as follows. Section II describes the proposed architecture including introduction to snow radar theory. In Section III, the radar system, hardware, and method are presented. Then, proof-of-concept measurements and discussion of results are reported in Section IV followed by conclusions in Section V.

II. Observation Principles

The primary constraints of a snow radar for long-term, unattended, autonomous snow observation are ultimately a tradeoff between functionality, performance, and cost. We present a first generation of a low-cost and low-power snow radar system with significant potential for improvement. The intention is to present a "proof-of-concept" and introduce a new method for *in situ*, high-quality snow assessment.

Aiming for a surface buried, upward looking radar system, low power consumption is important due to the potential interference with the covering snow pack. Unavoidably, overall power consumption will increase proportionally to the number of measurements. We previously presented a detailed power consumption analysis [21] of the instrument providing a road map for feasible improvements. However, this first generation is still considered "low power" and adequate for functional assessment. In addition to primary sensing functionality, the proposed radar instrument utilizes low-cost, standard consumer technology for simple wireless interconnect. In the following, we demonstrate the functional properties for snow pack assessment and outline future potential.

A. Snow Radar

The radar system is classified as an ultrawideband, coherent pulsed radar system. A pulsed radar measures the two-way-travel time (TwT) or time-of-flight (ToF) from the antenna to some reflective (backscattering) object or medium. TwT is used to measure the distance to the object, provided the propagation velocity is known (speed of light in air). Inversely, the propagation velocity may be estimated if the signal travel distance is known. The propagation velocity and signal amplitude of the received radio wave are characterized by the complex dielectric constant of the substance exposed and is defined as $\epsilon = \epsilon' - j\epsilon''$. However, dry snow will primarily affect the permittivity ϵ' (propagation speed). In this article, the permittivity ϵ' denotes the permittivity relative to vacuum (ϵ_0).

As radio waves penetrate a substance such as snow, signal propagation speed (permittivity) is reduced proportional to snow density. Measurable reflections occur with abrupt permittivity changes. Uniform dry snow give minor measurable reflections, while a sharp contrast like a layer of ice or the top of the snow cover (snow/air) would reflect a fraction of the signal detectable with radar. The amplitude of the signal reflected is proportional to the permittivity difference between layers in the snowpack as indicated by the reflection coefficient, Γ defined as [22]

$$\Gamma = \frac{\sqrt{\epsilon_1} - \sqrt{\epsilon_2}}{\sqrt{\epsilon_1} + \sqrt{\epsilon_2}} \tag{1}$$

where ϵ_1 is the permittivity of the first material and ϵ_2 is the permittivity of the second material [23].

Furthermore, the return loss can be expressed in either decibel as in (2), or in terms of percentage as in (3)

$$\mathrm{RL_{dB}} = 20\log_{10} \mid \Gamma \mid \tag{2}$$

$$\mathrm{RP_{\%}} = 100 \mid \Gamma \mid^2 . \tag{3}$$

Dry snow is a mixture of air and ice and permittivity varies with the volumetric fraction between air ($\epsilon' = 1$) and ice ($\epsilon' \approx 3.16$). However, the internal dielectric transitions in a uniform snow pack are very low and random resulting in minor radar reflections demanding very sensitive radar equipment.

The best case scenario would be a transition going from air to solid ice, which would result in a directional reflection of 8% of the exposed energy or $R_l = 11$ dB. This is the surface reflected energy, excluding free-space loss. Fortunately, such icy layers occur in snowpacks and provide important structural information visible with radio waves.

Surface scattering occurs with systematic permittivity differences like the border between snow and air. However, discrete particles can be present in the dielectric media causing volume scattering. The size of the particles relative to the wavelength determines the contribution of the volume scattering. Increased center frequency increases the snow scatter and backscatter coefficient, resulting in decreased penetration depth [23].

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In reality, the reflections occurring inside the snow would be much smaller, as the snow pack density is often gradual due to weight compression. These weak internal reflections make detailed radar imaging of snow stratigraphy very challenging. However, major density changes due to midwinter rain and melt gives sharp and easily detected reflections.

The top of the snow cover may be hard to identify during build-up, but varying weather conditions (like sun and temperature variations) will establish a visible reflector as reported in [14]. In addition, changes to the snow pack occur at the top, as snow accumulates, melts, or settles. These changes are traceable based on a time-lapsed measurement history.

Sensitive radar systems tend to be power consuming, expensive, and bulky. In this article, we explore a miniaturized, single-chip radar technology with high sensitivity providing the potential for remote and long-term snow-pack monitoring. To demonstrate feasibility, we evaluate our system for average SWE measurement of a snow pack and compared with manual snowpit assessment.

B. Snow Water Equivalence

SWE is considered as an important snow metric, giving an estimate of the total water depth contained in the snow. SWE is calculated from snow depth and snow density (ρ_s) [24]. SWE [25] is measured in meters of water and is defined as

$$\text{SWE} = h_s \frac{\rho_s}{\rho_w} \text{ [m]} \tag{4}$$

where h_s is the snow depth, ρ_s is the measured snow density [g/cm^3] and $\rho_w = 1$ (g/cm^3).

The dry snow density can be related to the permittivity (ϵ'). Dry snow is a mixture of ice and air and the permittivity depends on the mixing ratio. Ice has a constant density of $\rho_{\text{ice}} = 0.9167$ g/cm^3. The volume fraction v_i is the amount of ice contained in the snow and could be expressed as: $v_i = (\rho_s/\rho_{\text{ice}}) = (\rho_s/0.9167)$, where ρ_s is the snow density measured in g/cm^3. Thus, if the snow density is 0.25 g/cm^3, the volume fraction v_i is 0.27, which means that the snow is composed of ≈27% ice and ≈73% air.

Several empirical models are reported describing the relationship between density or volume fraction and the permittivity. The Tinga-Voss-Blossey formula is considered to be a good approximation for dry snow assessment [23]. Different permittivity models were compared in [26]. A frequently used model is proposed in [27]

$$\epsilon'_{\text{ds}} = \begin{cases} 1 + 1.4667v_i + 1.435v_i^3, & \text{for } 0 \leqslant v_i \leqslant 0.45 \\ (1 + 0.4759v_i)^3, & \text{for } v_i \geqslant 0.45. \end{cases} \tag{5}$$

The permittivity can be estimated for any transparent dielectric media with radio waves by measuring the TwT or velocity (V) through the media. ϵ' can be expressed in terms of TwT or velocity like this

$$\epsilon' = \left(\frac{c}{V}\right)^2 = \left(\frac{c \cdot \text{TwT}}{2R}\right)^2 \tag{6}$$

where c is the speed of light [m/s], V is the velocity [m/s], TwT is the two-way travel time [s], and R is the distance [m].

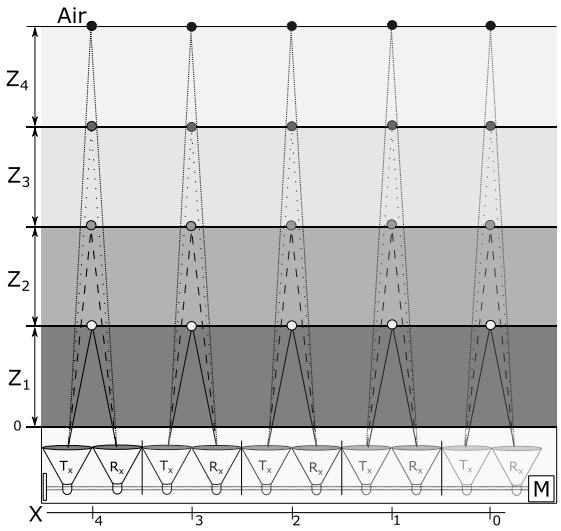


Fig. 1. Motor (M) drives one antenna pair consisting of one R_x and one T_x along a rail, illustrating the creation of stripmap SAR image.

Dry snow permittivity may be simplified to a first-order approximation as $\epsilon'_{\text{ds}} \approx 1+2\rho_s$ [28], for illustrations purposes. Results reported in this article are based on the full model described in (5). Thus, the average SWE also called bulk SWE for dry snowpacks could be approximated and expressed in terms of distance or permittivity as

$$\text{SWE}_{\text{bulk}} \approx \frac{(\text{TwT} \cdot c)^2 - 4R^2}{8R} \approx \frac{c \cdot \text{TwT}\left(\epsilon'_{\text{ds}} - 1\right)}{4\sqrt{\epsilon'_{\text{ds}}}}. \tag{7}$$

As shown in (7), SWE measurements require snow depth or permittivity in addition to TwT as indicated above for single (one-point) radar measurements.

Snow radar measurements are mostly carried out looking down toward the snow from a tower with known mounting height. A bistatic SAR radar system will capture TwT in several locations and does not require snow depth provided adequate snow surface reflections are received.

III. Imaging Radar Systems

The spatial resolution of an multi-static imaging radar system is limited by the number of stationary receivers (R_x) along the aperture. The number of receivers is limited by cost, power, and space constraints. Synthetic aperture radar (SAR) may synthesize a large aperture by systematically changing the distance between two radars doing repeated measurements with small increments. Given the medium of study is not changing during measurement period, a bistatic (two) radar system with one movable radar module creates new opportunities. The penalty is the additional complexity of a mechanical platform, expected to be acceptable with modern, light-weighted radar systems and discussed in [21].

A. Stripmap SAR

The simplest measurement scenario is a single, movable radar module traveling along the observation axis, often called stripmap SAR as shown in Fig. 1. Stripmap SAR gives

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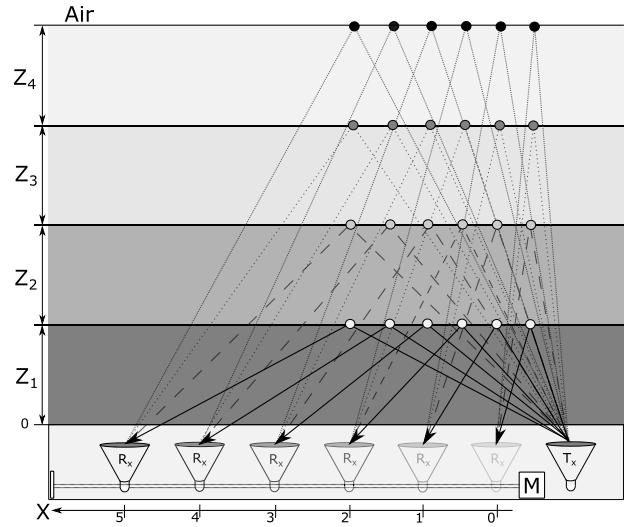


Fig. 2. Illustration of a upward looking radar conducting a bistatic normal move out measurement.

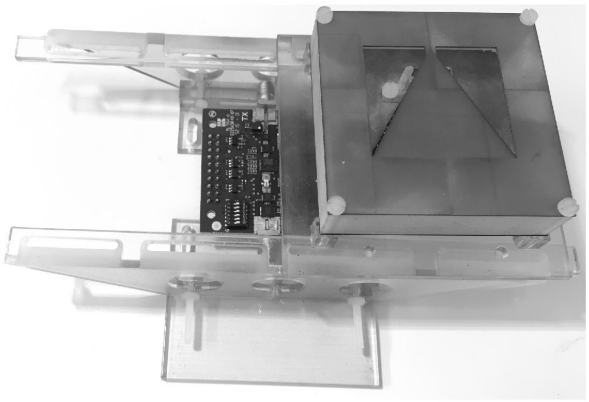


Fig. 3. Radar gantry with one antenna removed uncovering the radar module. The complete gantry measures 15.5×8 cm^2, and the patch antenna measures 6×6 cm^2.

an intensity image based on one or more reflections that are occurring from one or more snow layers. Such single-module systems require minor calibration and are simple to operate. However, stripmap SAR is unable to determine the propagation velocity through the medium, as no depth reference is available. Furthermore, SAR would not be able to measure the density for individual snow layers Z_1 to Z_4. Consequently, stripmap SAR only measures bulk snow indirectly as a homogeneous mass using additional information of snow depth or permittivity properties for determining SWE, as previously shown in (7). However, a downward looking radar mounted in a tower in fixed location would be able to measure bulk SWE directly, since the distance from ground to the radar is known. The term Stripmap SAR is in this article used for describing a radar system with both the receiver and transmitter moving along track. No SAR focusing algorithms are applied on the data in this article.

Although Stripmap SAR is not well suited for SWE measurement using upward looking radar, snow stratigraphy profiling is feasible.

B. Bistatic SAR

Bistatic SAR introduces velocity measurements by adding an additional fixed radar, as shown in Fig. 2 [29]. With bistatic SAR, high spatial resolution is possible with the added benefit of velocity measurements. The additional radar module adds some complexity and requires careful synchronization and calibration but provides the potential to characterize the full snow pack without any additional information (e.g., depth measurements).

The permittivity of different snow layers can be estimated using velocity picking [30] for each dominant reflection. We are also able to calculate SWE directly using (7), since both TwT and ϵ'_{ds} are known.

The functional relation between TwT and a linear change of distance between sender (T_x) and receiver (R_x) is hyperbolic providing different velocity measurements as radar module is moved. Assuming homogeneous and flat snow layers, a well-established estimation procedure called normal moveout (NMO) correction may be used for velocity (and permittivity) estimation. TwT as a function of module distance may be expressed as [30]

$$T(u) = \sqrt{T_0^2 + \frac{u^2}{V^2}} = \sqrt{T_0^2 + \frac{u^2 \cdot \epsilon'}{c^2}} \tag{8}$$

where T_0 is the travel time when the modules are closest, u is the position along the rail, ϵ' is the estimated permittivity, c is the speed of light, and V is the estimated velocity. The sampling interval ($\Delta_u = u_1 - u_0$) should satisfy the stripmap SAR sampling interval of ($\lambda/4$) [31] for mitigating grating lobes if SAR focusing is applied. Therefore, by "picking the velocity" making measurements line up along a reflecting surface, a correct velocity estimate is assumed. Bistatic radar combined with velocity picking estimates velocity for all measurable reflections and the density and SWE can be obtained for individual snow layers. In our measurements, we are using 30 different positions giving reasonably good statistical estimate of permittivity. However, the number of positions may be increased for better estimation.

Only the receiver antenna (R_x) is moved in Fig. 2, thus the measurement point changes over the aperture. As such, the bistatic radar may provide stratigraphy imaging of the snowpack. Although limited aperture, abrupt weather changes may still give interesting images provided both spatial and temporal resolution is adequate. In stable weather conditions, bulk SWE may be measured provided snow/air reflection is observable and homogeneous snowpack over the aperture is assumed. However, errors in the estimated velocity may occur because of local topography changes. These errors will decrease with increased snow depth as the reflection angle decreases. The same point could be observed by moving both the sender (T_x) and receiver (R_x) antenna symmetrically around a common midpoint, called common midpoint gathering or normal move out measurements (NMO). However, additional mechanical complexity and power consumption as well as no SAR imaging of the snow would be possible.

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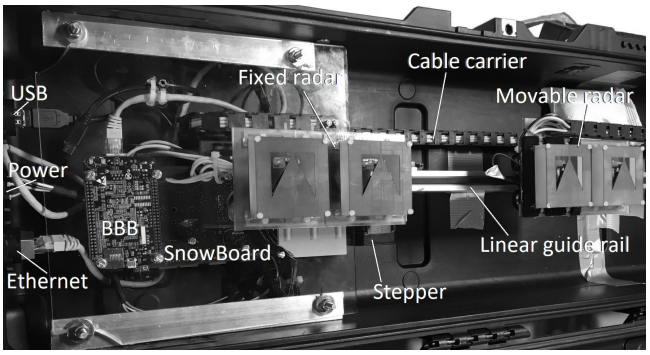


Fig. 4. Internals of the first Snowradar prototype.

C. Radar Module

The proposed radar system explores modern technology for building a power-efficient and low-cost, bistatic radar system for long-term, remote sensing. The central component of the snow-radar is the radar module, as shown in Fig. 3. The core component is a CMOS pulsed radar chip (X2) delivered by Novelda AS [32]. The pulsed, microwave radar system-on-chip (SoC) has a programmable center frequency from 5.3 to 10.2 GHz with a −10-dB bandwidth from 1.75 to 2.5 GHz. The active power consumption is <120 mW. The SoC is designed for unlicensed use (FCC-band) with Part15 limited output power (−41.3 dBm/MHz). For improved signal penetration, the center frequency has been reduced to 4 GHz with a bandwidth of 1.5 GHz, as the backscatter coefficient increases with frequency as given in Section II-A. Two lightweight patch antennas built with printed circuit boards (PCBs) and 3-D printed support are used for both R_x and T_x and are mounted directly on the radar RF PCB using SubMiniature version A (SMA) connectors.

D. Implementation

The radar system is encapsulated in a rugged polypropylene (PP) case designed to handle harsh outdoor conditions. The entire systems operate within the case. The length of the linear guide rail allows for 1 m of separation between the antennas. The largest antenna separation between the simulated phase center of the T_x and R_x antenna is 1.09 m, and the smallest is 9 cm. A photograph of the radar system is shown in Fig. 4. The presented measurements use 30 measurement points along the rail, with a along-track sampling interval (Δ_u) of 33.33 cm. The sampling interval is larger than ($\lambda/4$), however, no focusing is applied and with the homogeneity of the snowpack the likelihood of ambiguities are low. The somewhat undersampled measurements were traded for faster sampling speed but may be programmed as desired down to a $\Delta_u = 310\ \mu$m.

The case is reasonably transparent to microwave, as polypropylen has a permittivity of ϵ_{pp} of ≈2.2. Conveniently, the air → polypropylen interface will cause a reflection from a known distance, useful for calibration purposes.

The first prototype is based around a single board computer called Beagle Bone Black. Additional support circuitry for positioning, power, and data acquisition is built with off-the-shelf components. The hardware and system architecture with power optimization scheme has previously been reported in [21].

E. Calibration

For proper operation, calibration is required since TwT is in the order of nanoseconds, while the incremental change of TwT between measurement positions is in the order of ≈75 ps.

Calibration starts by moving the radars next to each other. The R_x antenna is now located in center, with one T_x antenna at each side. Consequently, the geometry is approximately equal for both the T_x antennas with respect to the R_x antenna and both radar modules are transceivers. The lid is assembled parallel to the linear guide and used as a reference reflector. By symmetry, the lid reflection from the T_x antenna will arrive at the same time for both the R_x antennas. Each radar system has programmable observation frame offset in steps of 24 ps used for very accurate timing alignment. The frame offset register of the movable radar is kept fixed, while the frame offset of the stationary radar is swept. The frame offset with the highest correlation for the lid reflection is used during snow measurements. The measurement for both radars with the highest correlation is saved, and the amplitude difference is calculated and used for amplitude correction in post processing.

This procedure calibrates the entire system chain including cable lengths, cable losses, antenna gain, and the radar receivers. Due to variable weather conditions resulting in temperature variations, calibration is carried out before each measurement for better precision but may not be necessary.

Previously reported upward looking snow radars have conventionally used a hoisting device for calibration and clutter removal in two different positions [14]. In this radar, instrument clutter is removed using horizontal movements. However, clutter removal is improved by averaged clutter removal from 30 different positions. When the horizontal distance between R_x and T_x antenna is increased, the travel time to reflections is also increased, while static clutter is unchanged, similar to decreasing the vertical height as in [14].

IV. Proof-of-Concept Measurements

The radar system is intended for continuous upward looking measurements. Our ambition was to present long-term, *in situ* measurement results from winter season 2017/2018 at Finse in Norway based on hourly reports. Unfortunately, the radar instrument was flooded and had to be rebuild. The radar was brought for the NASA SnowEX 2017 campaign at Grand Mesa, Colorado, United States of America (USA) and could not be buried underneath the snow without destructive interference. In order to provide some indications of proper performance, we present initial proof-of-concept measurements collected looking downward and compared to a manual inspection of the assessed snowpack.

A. Reference Measurements

A traditional snow pit survey was conducted by a trained inspector at the radar site for comparison and observed results

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TABLE I
MANUALLY OBSERVED SNOW PIT DATA OF JC6 LSOS 2017-02-15 10:15

Density				Stratigraphy					
Height [cm]		Profile A	Profile B	Height [cm]		Grain Size [mm]			Grain
Top	Bot.	[kg/m^3]	[kg/m^3]	Top	Bot.	min	max	mean	Type
96	86	117	108	96	90	0.4	1	0.7	M
86	76	185	174	90	78	0.6	1.3	0.9	M
76	66	254	252	78	76	1.5	2.5	1.8	MF
66	56	282	286	76	56	1	1.5	1.3	MF
56	46	296	292	56	40	1.2	2	1.6	MF
46	36	339	329	40	35	1.6	2.5	2.3	MF
36	26	333	332	35	31	2	3.5	3	MF
26	16	359	343	31	25	1.4	2.2	2	MF
16	6	361	346	25	12	1.3	2.3	1.8	MF
Total Depth : 96 cm				12	4	1.3	2.5	2	MF
S: Dr. Ludovic Brucker NASA/GSFC				4	0	NA	NA	NA	SLUSH

Where M is Melt, and MF is Melt Freeze.

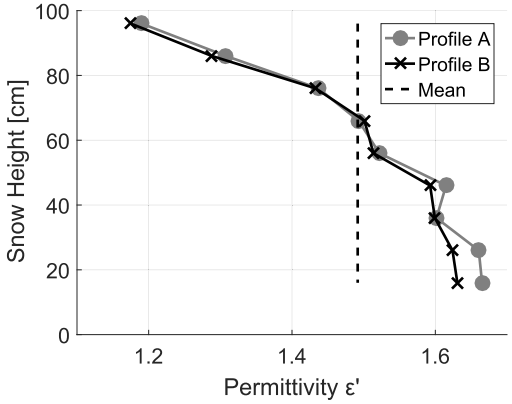


Fig. 5. Permittivity calculated for the snow pack presented in Table I using (5).

are given in Table I. The survey was done immediately following the radar measurements.

The dry snow permittivity is calculated from density using (5) and plotted for the snowpack in Fig. 5. As expected, the permittivity increases gradually with snow depth due to weight compression. The pack is relatively homogeneous with no high contrast ice or liquid water layers. Automated retrieval of stratigraphy is not attempted within this analysis. However, a layer of wet snow in the bottom provides adequate signal reflection for bulk SWE measurements.

SWE can be calculated from the density given in Table I using (4). The bulk SWE is obtained by averaging density throughout the pack for both profiles, thus $\rho_{s_{\text{ref}}} = 0.27$, $\epsilon'_{\text{ref}} = 1.49$, and SWE $= 0.25$ (excluding the layer of slush) assuming dry snow.

We get excellent signal penetration through the snowpack with a clear strong reflection as shown in Fig. 7(a) with an attenuation of only 7.337 dB over a distance of 172 cm. Unfortunately, moisture or temperature was not measured and not reported in Table I. However, minor signal attenuation indicate low moisture. In [14], wet snow is reported with $\epsilon' > 2$ significantly higher than measured for the observed snowpack indicating that dry snow approximation is adequate.

Slush is defined as snow with a liquid water content $>5\%$ [33]. The moisture content was not measured in the pit. However, snow with a volumetric moisture content of 5% and a density of $\rho_s = 0.28$ yields a dielectric constant of $2 - j0.21$ [20] at 4 GHz due to dielectric loss. The pit was

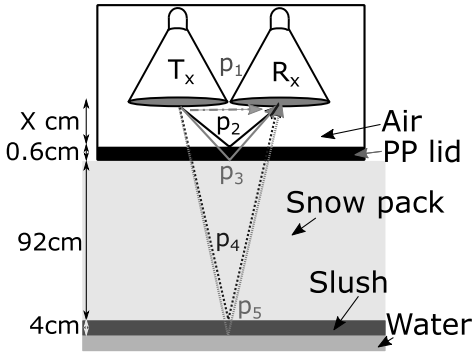


Fig. 6. Simplified measuring model.

TABLE II
ASSUMED PROPAGATION PATHS FOR FIG. 6 WITH ESTIMATED REFLECTION COEFFICIENTS

Path	Description	$\lvert\Gamma\rvert$	RP %
p_1	Direct path	-	-
p_2	Air $\rightarrow$ Radar enclosure lid	0.19	3.8
p_3	Radar enclosure lid $\rightarrow$ Light snow	0.15	2.42
p_4	Snow $\rightarrow$ Slush	0.145	2.15
p_5	Slush $\rightarrow$ Water	0.67	47

located over a creek, so the liquid water content is higher further down toward the creek bed, and increasing the moisture content to 10% yields $2.8 - j0.42$. Finally, pure water has a dielectric constant of $74.3 - j30.8$ at 4 GHz. This provides a distinct boundary condition for our analysis.

B. Measurement Analysis

Inspecting the manual report in Table I, there are likely five dominant energy and/or propagation paths for this measurement indicated in Fig. 6. The first path p_1 is the direct path between the antennas, p_2 is the reflection path from the air to the Polypropylen lid of the radar enclosure, p_3 is the enclosure lid $\rightarrow$ snow reflection path, p_4 is the snow $\rightarrow$ slush reflection, and p_5 is the slush $\rightarrow$ water reflection. The reflection coefficients for the different reflections are estimated in Table II.

The down range resolution is given by the bandwidth B_w for monostatic radar [Hz], and defined as $\Delta R = (c/2B_w)$ [31], thus for this system $\Delta R = 0.06$ m. Additional resolution refinement is possible exploring coherency of the reflected signal as well as penetration speed reduction as a function of permittivity. Consequently, the first reflection will be a sum containing the first three reflections and/or paths, namely, p_1, p_2, and p_3. The second reflection will be a product consisting of both the slush reflection p_4 and the water reflection p_5. We predict two strong reflections, given the pit data in Table I, later verified by measurements.

C. Monostatic Bulk SWE Measurement

We assume a start reflection and a stop reflection as described in Section IV-B. The TwT between these two reflections is the two-way propagation time through the snow pack from the top of the snow pack down to the slush layer. Furthermore, the depth is known from the pit-sheet in Table I,

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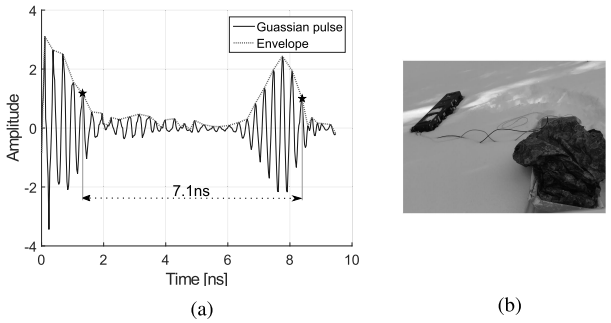


Fig. 7. Monostatic measurement. (a) Raw monostatic data showing the two-way mean travel time of 7.1 ns from lid to slush. (b) Actual measurement setup.

and thus the bulk permittivity can be measured through the snow pack using (6).

The measured pulse is plotted in Fig. 7(a). As may be seen, the radar system is fully coherent providing both the amplitude and the phase of the received Gaussian shaped carrier signal. The transmitted bandwidth is given by the envelope, but the phase changes may be explored for improved depth resolution. The time difference between the two pulses can be measured at several different locations of the Gaussian pulse, peak or zero crossing are often used. The end of the pulse has a well recognizable form, which is used to measure the TwT = 7.1 ns.

The depth of the snow pack is estimated to 96 cm. However, the slush layer is 4 cm, and the thickness of the lid is 0.63 cm, thus the distance R should be $R = 91.4$ cm. Using (6) yields an $\epsilon' = 1.34$ with a $\rho_s = 0.17$ for the monostatic measurements, which is lower than manual inspection results yielding a mean $\rho_s = 0.25$. The low ϵ' value obtained by the monostatic measurements is probably due to inaccurate snow depth/height combined with added snow compression from instrument weight [see Fig. 7(b)]. Thus, by correcting the distance of 5 cm for the monostatic measurement, yields an $\epsilon' = 1.5$ with a $\rho_s = 0.25$. However, bistatic measurements are expected to be more accurate since no snow depth estimation is required.

D. Bistatic SWE Measurement

Initial static clutter removal as shown in [34] is applied. The TwT estimate in a given position is extracted from the Gaussian radar signal using a Hilbert transform for envelope detection as shown in Fig. 7(a). Normal move-out correction (NMO) is used for permittivity estimation as described in Section III-B. The first large reflection shown in Fig. 8 up to 1 ns originates from lid reflections used as start reference for snow cover. The TwT through the snow pack starts ≈1 ns and reflections from the slush layer appear ≈7.5 ns later. The permittivity is estimated using NMO and best fit gives $\epsilon' \approx 1.5$ and illustrated in Fig. 8.

The carriage starts to the right of the plot in Fig. 8 at the 1 m mark and moves left toward 0 m. With modules located as close as possible (0 m), the signal path is shortest with lowest TwT. The first reflection from the lid of the box is stronger

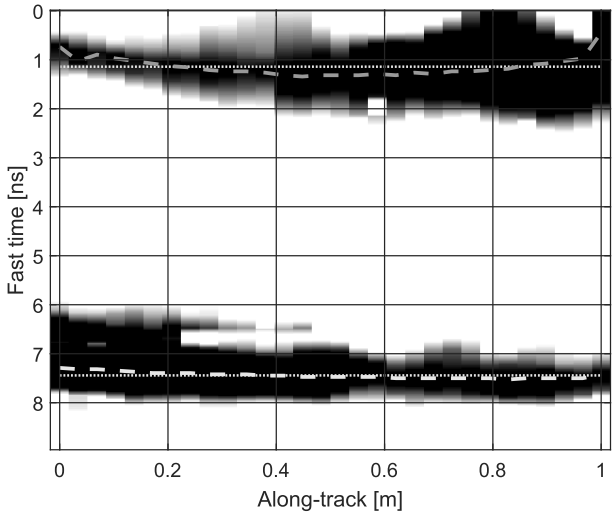


Fig. 8. Bistatic envelope measurement with manual velocity picking. The first reflection is corrected for air ($\epsilon' = 1.$) and the second reflection is corrected for an $\epsilon'_s = 1.5$. Thick dashed line: peak pulse. Thin dashed line: mean. Data set: JC6 2017-02-15 10:30:53. Maximum value is 1 indicated by black, while white is 0.

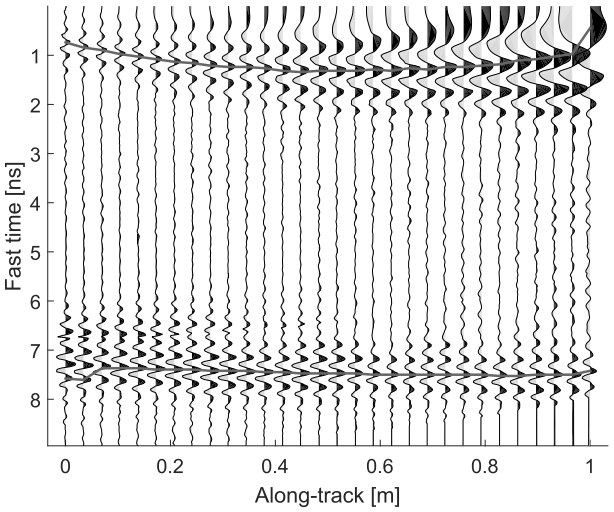


Fig. 9. Bistatic measurement with manual velocity picking of data set JC6 2017-02-15 10:30:53 shown as a wiggle plot, with the peak of both the reflections traced.

both due to better incident angle and shortest traveling time. Furthermore, the low incident angle occurs for the slush/snow as well giving an improved radar image. Less detail is captured when the along-track distance increases, as the incidence angle increases toward the brim of the antenna beam.

A "wiggle" plot of the same data plotted with the UWB pulse is shown in Fig. 9 where the carrier phase information provides more details. Tracing the strongest peak improve the precision significantly as indicated by the thick dashed line in Fig. 8.

Further refinement is possible by interferometry. The first measurement of the matrix S at 0 m are complex conjugated with subsequent 29 measurements as

$$S(n) = S(n+1) \circledast S(1)^*, \quad n = 1, 2, 3, \ldots, 29. \tag{9}$$

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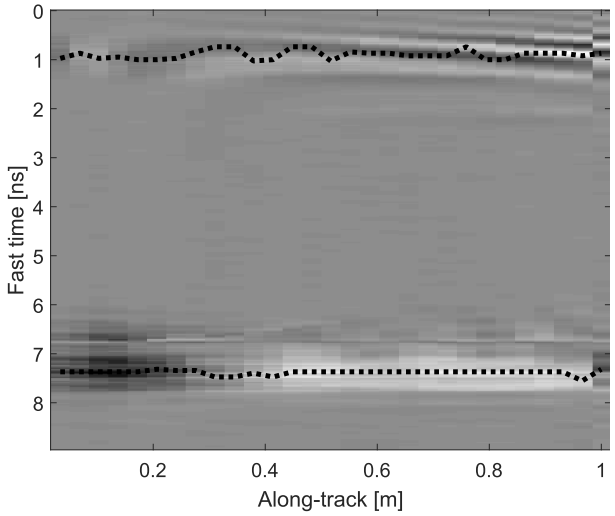


Fig. 10. Phase component plotted for data set JC6 2017-02-15 10:30:53, with the phase component of the first measurement subtracted from the remaining 29 along track measurements. Dashed lines mark peak pulse of the first and second reflection. Maximum value is 1 indicated by white, while black is 0.

TABLE III
AUTOMATED BULK VELOCITY ESTIMATES

Data set	ϵ_s'	$\rho_s \left[\frac{g}{cm^3}\right]$	**R** [m]	**SWE** [m]
2017-02-15-09-31-32	1.6	0.33	0.85	0.28
2017-02-15-10-24-14	1.45	0.26	0.8	0.21
2017-02-15-10-30-53	1.45	0.26	0.8	0.21
2017-02-15-10-38-29	1.5	0.28	0.86	0.24
2017-02-15-11-04-30	1.6	0.33	0.83	0.27

The phase difference between the first measurement and the 29 remaining measurements are plotted in Fig. 10 as $\mathrm{Im}\{S\}$. We can see that using phase change or interferometry further improves the detection of the two layers of interest. We expect further signal processing may disclose more details of snow pack stratigraphy, but this is beyond the scope of this analysis.

As we are looking down into a creek with a distinct slush layer, we have a dominant reflection from the snow $\rightarrow$ slush-interface. Propagation velocity (v) in (8) is swept in values corresponding to velocities in media with a ϵ' of 1 to 2, with 0.05 step interval. The second, the dominant slush/water reflection, is illuminated by applying time gating. The NMO is calculated, and summed for each of the swept velocities, correct bulk velocity occurs when the measurements are aligned in time and the sum is maximized. Thus, the measurements are correctly "stacked" [30].

Five data sets were collected over a duration of 93 min. The first measurement 09-31-32 was taken at the same transect as the pit before it was dug. The subsequent measurements were taken after the pit was completed. Measurements 10-24-14 and 10-30-53 were taken normal to the pit wall. Measurements 10-38-29 and 11-04-30 were taken parallel to the pit wall with in a distance of 1.5 m. The previously described automated bulk estimate approach was applied and the results are given in Table III.

With only five radar measurements, the statistical basis is limited. However, with the data at hand, we achieve a mean bulk SWE $= 0.24$ m with a standard deviation of $\sigma = 0.033$. The mean SWE obtained by the bistatic radar of 0.24 m is comparable to the SWE of 0.25 m measured using conventional snow pit inspection as given in Table I. Furthermore, the bistatic radar configuration offers the potential of improved snow assessment as well as radar profiling of the snow cover.

The obtained measurement precision of SWE is 15% due to few measurements and is expected to improve with proper validation.

Density measurements using traditional density cutters have an approximate measurement uncertainty of 9%. We get a density measurement precision within 13% of the mean somewhat more than 9% measurement uncertainty of traditional cutters. However, increased number of measurements in controlled environment is required for proper statistical validation.

V. CONCLUSION

In this article, we present a novel snow assessment instrument exploring modern low-cost, low-energy radar modules. The proposed radar system is capable of determining bulk SWE without any additional observations exploring natural reflective snow layers as well as creating time-lapsed, snow profiles for long-term, remote snow monitoring. Initial average or bulk SWE radar measurements agree well with manual snow pit observations. Although empirical qualification is limited, we believe these initial assessments indicate a power-efficient, upward looking, bistatic radar instrument for unattended, remote snow monitoring is feasible. Hopefully, long-term details of snow pack behavior may give insight of scientific significance. Potentially, the new bistatic SAR instrument may be used for both dry and wet snow assessment and provide detailed stratigraphic snow imaging of significance.

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